existing proprietary data feeds that contain this information to competing consolidators and self-aggregators.[1218] These data feeds may contain information that goes beyond what is necessary to generate consolidated market data, but competing consolidators and self-aggregators will not be billed based upon the data feed that they receive. Similar to the current billing, reporting and audit processes, the administrator of the effective national market system plan(s) could be expected to license and bill and, when required, employ an audit process to assess the usage of the data content made available to competing consolidators and self-aggregators under Rule 603(b) for billing purposes.[1219]

(h) Comments on Timestamps

As proposed, Rule 614(e)(1)(ii) required that the amendment to the effective national market system plan(s) for NMS stocks include provisions requiring the application of timestamps by the SRO participants on all consolidated market data, at the time the consolidated market data component was generated by the SRO participant and at the time the SRO participant made the consolidated market data available to competing consolidators and self-aggregators.[1220]

1218 See supra Section III.B.9(b).

1219 See supra Section III.E.2(e).

1220 The SROs currently submit timestamped data under the Equity Data Plans and the National Market System Plan Governing the Consolidated Audit Trail ("CAT NMS Plan"). See, e.g., CTA Plan, supra note 10, at Section VI.(c); Nasdaq UTP Plan, supra note 10, at Section VIII; CAT NMS Plan at Sections 6.3(d), 6.8, available at https://www.catnmsplan.com/sites/default/files/2020-02/CAT-2.0-Consolidated-Audit-Trail-LLC%20Plan-Executed_%28175745081%29_%281%29.pdf (last accessed Nov. 27, 2020). See also 17 CFR 242.613; Securities Exchange Act Release No. 78318 (Nov. 15, 2016), 81 FR 84696, (Nov. 23, 2016) ("CAT NMS Plan Approval Order"). The CAT NMS Plan was Exhibit A to the CAT NMS Plan Approval Order. However, the limited liability company agreement of a new limited liability company named Consolidated

One commenter stated that the proposal would require the effective national market system plan(s) participants to apply timestamps to consolidated market data even though they were not consolidating and disseminating consolidated market data.[1221] The Commission has modified the language of Rule 614(e)(1)(ii) to require the SRO participants to apply timestamps to all information with respect to quotations for and transactions in NMS stocks that is necessary to generate consolidated market data and not to consolidated market data as the proposed rule required. Specifically, the timestamps applied by the SROs must be to the individual components of data content underlying consolidated market data, i.e., all of the individual components of data content underlying core data, regulatory data, administrative data, self-regulatory organization-specific program data, and additional elements defined as "consolidated market data."

This commenter also criticized the proposal for underestimating the burdens of adding timestamps.[1222] The Commission disagrees with the commenter regarding the burden of adding timestamps. The SROs currently add timestamps to all elements of consolidated market data and thus, the Commission does not believe that ensuring that timestamps are applied in a consistent manner going forward would impose significant, if any, costs to the SROs. Timestamps are important for market participants as they provide the ability to measure latency and ensure accurate sequencing of data. The application of timestamps may also incentivize the SROs to

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Audit Trail, LLC now serves as the CAT NMS Plan. See Securities Exchange Act Release No. 87149 (Sept. 27, 2019), 84 FR 52905 (Oct. 3, 2019).

1221 See NYSE Letter II at 21.

1222 See id.

make available their consolidated market data as quickly as possible. Therefore, the application of timestamps needs to be consistent and reliable.[1223]

(i) Comments on Annual Assessment

As proposed, Rule 614(e)(1)(iii) required the amendment to the effective national market system plan(s) for NMS stocks to reflect that the participants are required to conduct an annual assessment of the overall performance of competing consolidators—including speed, reliability, and cost of data provision—and provide the Commission with a report of such assessment on an annual basis. The Equity Data Plans play an important role in governing the operation of the national market system. The Commission believes that the effective national market system plan(s) for NMS stocks should continue in this important role by monitoring the overall performance of the provision of consolidated market data by competing consolidators to seek to ensure that the decentralized consolidation model is operating soundly and is therefore adopting this provision, as proposed, with one modification.

As described in the Proposing Release, the plan must assess several key factors of the operation of the decentralized consolidation model, including: (1) the speed of competing consolidators in receiving, calculating, and disseminating consolidated market data products; (2) the reliability of the transmission of consolidated market data products; and (3) a detailed cost analysis of the provision of consolidated market data products. The effective national market

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1223 SRO timestamps will also assist market participants in their ability to assess latencies in the provision of consolidated market data. Under Rule 614(d)(3), competing consolidators are required to make available consolidated market data products that include timestamps assigned by the SROs as well as competing consolidators. Competing consolidators will be required to timestamp the data underlying consolidated market data at specific intervals: (1) upon receipt from an SRO at the SRO data center, (2) upon receipt at its aggregation mechanism, and (3) upon dissemination of consolidated market data to customers. See supra Section III.C.8(a) and the discussion of Rule 614(d)(4).

system plan(s) would base their assessments on the information made publicly available by competing consolidators, including the information that each competing consolidator is required to make available under Rule 614.

One commenter supported requiring the filing of a proposed plan amendment to mandate an annual assessment and suggested that the annual assessment be made public to further assist broker-dealers in selecting competing consolidators.[1224] One commenter stated that "the Proposal does not indicate whether the results of particular assessments will be made publicly available to firms and what, if any, actions broker-dealers will be required to make in response to such assessments."[1225] One commenter suggested that the annual report not review individual competing consolidator performance "in silo" by also reviewing at the competition.[1226]

The Commission is adopting the rule, with the addition that the annual report would be made publicly available by the Commission. The Commission believes that the annual report should be made publicly available to provide transparency to investors as to the operation of the national market system. The Commission believes that the annual report can assist the Commission in monitoring and evaluating the operation of the national market system and decentralized consolidation model. The annual report, however, is not an assessment of individual competing consolidators but of the overall performance of the provision of

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1224 See Clearpool Letter at 9.

1225 TD Ameritrade Letter at 13. This commenter asked several questions about the expectations on broker-dealers in response to the annual report. See id. at 13–14. As discussed above, the annual report will not be a report on individual competing consolidators but rather a report on the operational status of the whole decentralized consolidation model.

1226 See Data Boiler Letter I at 64 ("How CC beat their competition among peers, and the overall industry rely less on Exchanges' PP and SAs' services are the best key performance indicators (KPIs).").

consolidated market data by competing consolidators. Market participants that want to evaluate the individual performance of a competing consolidator can utilize the individual competing consolidator's disclosures on its Form CC and the monthly performance metrics published by each competing consolidator.

Another commenter stated that the SROs would incur costs associated with assessing competing consolidators although the effective national market system plan(s) would not have a role in selecting or monitoring competing consolidators.[1227] The SROs currently incur costs in overseeing the national market system and some of these costs may change in the decentralized consolidation model, including the new costs associated with conducting an assessment and developing the annual report. The Commission does not believe that the costs should be overly burdensome. As stated above, the Operating Committee can use public reports of competing consolidator performance as well as any pertinent information that the plan(s) believe would be useful to assess competing consolidators and develop the annual report. Further, the Commission believes that the Operating Committee is well-suited to perform this assessment. The SROs will be making the data content underlying consolidated market data available to competing consolidators and establishing the necessary connectivity to competing consolidators, and as stated, the effective national market system plan(s) will continue to have important responsibilities in developing, operating, and regulating the national market system. The Commission believes that the Operating Committee should develop the annual report as a means to monitor the overall performance of competing consolidators and to seek to ensure that the national market system is operating soundly.

[1227] See NYSE Letter II at 28.

(j) Comments on List of Primary Listing Exchanges

Finally, proposed Rule 614(e)(1)(iv) required the amendment to the effective national market system plan(s) for NMS stocks to include a list of the primary listing exchanges for each NMS stock.[1228] The primary listing exchanges will be required to collect, calculate, and make available regulatory data to competing consolidators and self-aggregators. Therefore, each primary listing exchange must be identified to determine who is responsible for collecting, calculating, and making regulatory data available. One commenter agreed with developing a list identifying the primary listing exchange.[1229] One commenter suggested that the Commission develop this list.[1230] The Commission believes that the plan(s) are best suited to develop the list and to ensure that it is kept current and readily accessible. The Commission is modifying the language of Rule 614(e)(1)(iv) to require that the plan(s) develop, maintain, and publish the list. The Commission believes that the list of primary listing exchanges should be maintained and published so that market participants will know which exchange is responsible for providing regulatory data. Further, competing consolidators and self-aggregators will need to know which exchange will be making regulatory data available.

(k) Regulation SCI

The Commission is modifying Rule 614(e) to accommodate the new definition of SCI competing consolidator classification under Regulation SCI. Specifically, new paragraph (v) of Rule 614(e) will require the participants to the effective national market system plan(s) for NMS stocks to file with the Commission an amendment that requires the plan(s) to calculate and

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1228 The term "primary listing exchange" is defined in Rule 600(b)(68).

1229 See Data Boiler Letter I at 64.

1230 See NYSE Letter II at 28.

publish on a monthly basis the consolidated market data gross revenues for NMS stocks as specified by: (1) listed on the NYSE; (2) listed on Nasdaq; and (3) listed on exchanges other than NYSE or Nasdaq. The Commission believes that the plan(s) are best suited to calculate and publish this information because, as noted above, the effective national market system plan(s) will charge the fees for the data content underlying consolidated market data, collect the revenue, and oversee accounts and billing. Competing consolidators will use the calculation and publication of consolidated market data gross revenues to assess whether they have reached the 5% threshold described in Rule 1000 for SCI competing consolidators. As discussed below, the Commission believes that competing consolidators that reach these thresholds should be held to higher systems resiliency and integrity standards as required under Regulation SCI than competing consolidators that are below this threshold.[1231]

F. Systems Capability: Amendment to Rule 1000 of Regulation SCI to Expand "SCI Entities" Definition to Include "SCI Competing Consolidator"; Adoption of Rule 614(d)(9): Systems Integrity

In the Proposing Release, the Commission stated its preliminary belief that competing consolidators should be subject to the requirements of Regulation SCI.[1232] The Commission adopted Regulation SCI in November 2014 to strengthen the technology infrastructure of the U.S. securities markets, reduce the occurrence of systems issues in those markets, improve their resiliency when technological issues arise, and establish an updated and formalized regulatory

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1231 See infra Section III.F.

1232 See Proposing Release, 85 FR at 16785–89.

framework, thereby helping to ensure more effective Commission oversight of such systems.[1233] The key market participants that are currently subject to Regulation SCI are called "SCI entities" and include certain SROs (including stock and options exchanges, registered clearing agencies, FINRA, and the Municipal Securities Regulatory Board) ("SCI SROs"); alternative trading systems that trade NMS and non-NMS stocks exceeding specified volume thresholds ("SCI ATSs"); the exclusive SIPs ("plan processors"); and certain exempt clearing agencies.[1234]

As the Commission stated in the Proposing Release, competing consolidators, as sources of consolidated market data, would serve an important role in the national market system. The Commission explained that, as it had stated when adopting Regulation SCI, "both consolidated and proprietary market data systems are widely used and relied upon by a broad array of market participants, including institutional investors, to make trading decisions, and … if a consolidated or a proprietary market data feed became unavailable or otherwise unreliable, it could have a significant impact on the trading of the securities to which it pertains, and could interfere with the maintenance of fair and orderly markets."[1235] For these reasons, Regulation SCI applies to both the exclusive providers of consolidated market data (i.e., the plan processors) and to proprietary market data systems, and is not limited to applicable systems of plan processors, but rather also includes the market data systems of any SCI entity, including SCI SROs. Taking into

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1233 See Securities Exchange Act Release No. 73639 (Nov. 19, 2014), 79 FR 72252 (Dec. 5, 2014) ("SCI Adopting Release"), at 72252–56 for a discussion of the background of Regulation SCI.

1234 See Rule 1000.

1235 See Proposing Release, 85 FR at 16786 (quoting SCI Adopting Release, supra note 1233, at 72275).

consideration the role of competing consolidators as providers of consolidated market data feeds that are likely to be widely used and relied upon by market participants, the Commission proposed to apply Regulation SCI to competing consolidators by including them within the definition of "SCI entity" and requested public comment.[1236] In particular, among other things, the Commission requested comment on whether all of the obligations set forth in Regulation SCI should apply to competing consolidators or whether only certain requirements should be imposed, such as those requiring written policies and procedures, notification of systems problems, business continuity and disaster recovery testing (including testing with participants/subscribers of a competing consolidator), and penetration testing.[1237]

A number of commenters supported applying the requirements of Regulation SCI to competing consolidators in some form.[1238] In particular, a few commenters supported application of Regulation SCI to competing consolidators as proposed.[1239] Others argued that competing consolidators should be considered to have "critical SCI systems" like the exclusive

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1236 In addition, the Commission proposed to revise the definition of "critical SCI system" to account, among other things, for the systems of OPRA's plan processor, since the competing consolidator model will not apply with respect to trading in options. See Proposing Release, 85 FR at 16786–87. The Commission is adopting the revision to the definition of "critical SCI system" as proposed. See infra notes 1315–1316 and accompanying text.

1237 See Proposing Release, 85 FR at 16789.

1238 See Cboe Letter at 26; Nasdaq Letter IV at 35–36; Data Boiler Letter I at 57; STANY Letter II at 6; FINRA Letter at 4, n.14; MEMX Letter at 8; Fidelity Letter at 3, 10; Clearpool Letter at 9.

1239 See FINRA Letter at 4, n.14; MEMX Letter at 8; Fidelity Letter at 3, 10. See also Clearpool Letter at 9; STANY Letter II at 6.

SIPs and thus subject to higher requirements than proposed.[1240] Some commenters, however, expressed concern that the costs of SCI compliance would be a barrier to entry and could deter entities from seeking to become competing consolidators.[1241] Similarly, several commenters, although not citing Regulation SCI specifically, expressed general skepticism about the ability to attract new entrants to register as competing consolidators, citing among other factors, potential lack of economic incentives.[1242]

The Commission continues to believe that competing consolidators, as providers of consolidated market data products, will serve an important role in the national market system. Thus, consistent with the views of many commenters, the Commission believes that it is important to impose requirements to help ensure that the technology systems of competing consolidators are reliable and resilient, consistent with the policy goals of Regulation SCI.[1243] The Commission is cognizant that Regulation SCI entails compliance burdens for new entrants[1244] and, in particular, that those costs could serve as a barrier to entry for potential

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1240 See Cboe Letter at 26; Nasdaq Letter IV at 35–36; Data Boiler Letter I at 57; STANY Letter II at 6.

1241 See NYSE Letter II at 15; ACTIV Financial Letter at 2; IDS Letter I at 13; STANY Letter II at 6–7; Angel Letter at 19–21. See also TD Ameritrade Letter at 13; Nasdaq Letter III at 4.

1242 See also supra notes 626–629 and 641–649 and accompanying text (discussing commenters' views that a lack of sufficient economic incentives for potential competing consolidators and the costs to become a competing consolidator outweigh the benefits).

1243 See supra note 1233 and accompanying text; Cboe Letter at 26; Nasdaq Letter IV at 35–36; Clearpool Letter at 9; Data Boiler Letter I at 57; FINRA Letter at 4, n.14; MEMX Letter at 8; Fidelity Letter at 10. See also IntelligentCross Letter at 5, BlackRock Letter at 5; ACS Execution Services Letter at 5; Temple University Letter at 1–2.

1244 See Proposing Release, 85 FR at 16808–09, 16836–38, 16845–48 (discussing paperwork burdens, costs, and benefits of complying with Regulation SCI). See also Nasdaq Letter